PSG Institute of Technology and Applied Research

Department of Artificial Intelligence and Data Science

Event-Driven Skill-Phase Estimation for Robot Manipulation Human Video-Based Skill Learning with Spiking Neural Networks

Submitted by :

Rahul A
Ragulnath P
Ranjith V
Nishanth P

Guided by :

Ms.R.S. Niranjana

Problem Statement

Traditional robotic arms require manually programmed trajectories for every manipulation task.

Existing limitations

- Robot trajectories are often **task-specific and manually programmed**.
- A human demonstration contains useful **motion, interaction and temporal information**, but extracting this information automatically is challenging.
- Human and robot embodiments differ, requiring **skill representation and motion retargeting**.
- During execution, the robot needs to know its **current phase/state of the learned task**.
- Conventional state-estimation approaches can introduce additional computational overhead.

Proposed Problem:

Develop a Learning-from-Demonstration framework that extracts manipulation skills from human RGB videos, learns a transferable skill representation, and uses an SNN-based temporal state estimator for phase-aware robotic execution.

Abstract

The proposed system aims to enable a robotic arm to learn manipulation skills from human demonstrations rather than relying entirely on manually programmed trajectories.

A human RGB video is first processed to extract **human motion, hand state and object interaction information**. The demonstration is temporally segmented into meaningful phases such as **Reach, Approach, Grasp, Lift, Transport, Place and Release**.The extracted demonstration is then represented as a transferable skill using **Learning from Demonstration (LfD)** techniques. The learned skill will subsequently be mapped to the robotic arm through trajectory generation and inverse kinematics.

During robot execution, an **Spiking Neural Network (SNN)** will act as a temporal state/phase estimator, identifying the robot's current phase of the learned manipulation skill.

Video → Skill → Robot Trajectory → SNN State Estimation → Phase-Aware Execution

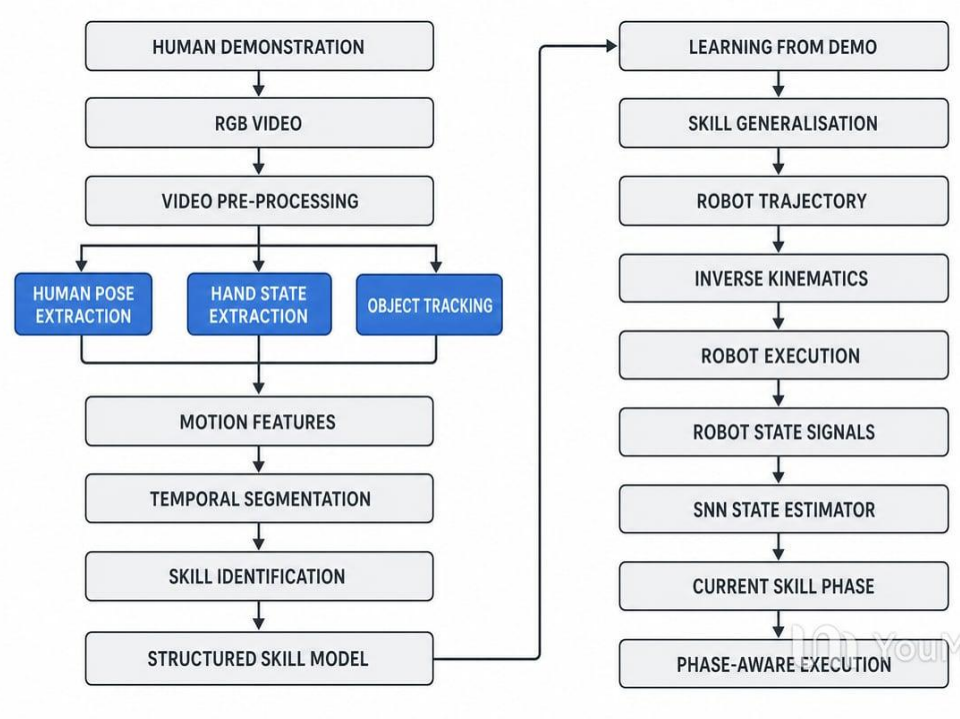
Literature Survey - I

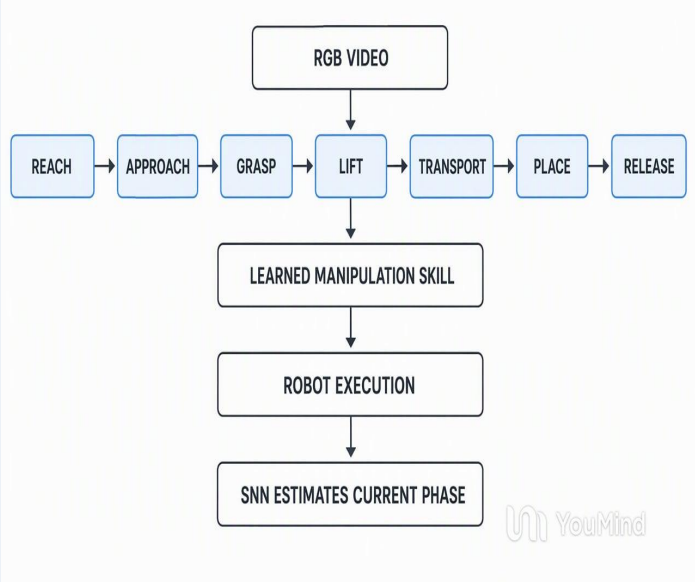
No.	Paper	Authors	Publication	Relevance
1	Robots Acquire Manipulation Skills in Seconds from a Single Human Video (HOST)	Guangyan Chen et al.	2026	Estimates robot progress within a demonstrated task and uses a shared task-progress manifold. Closest to our phase concept.
2	Video2Sim2Real: Full-Stack Autonomous Dexterous Skill Acquisition from a Single Human Video	Yunhai Han et al.	2026	End-to-end skill acquisition from human video with simulation, IL, residual RL and real-world execution.
3	Vision-based Manipulation from Single Human Video with Open-World Object Graphs (ORION)	Yifeng Zhu, Arisrei Lim, Peter Stone, Yuke Zhu	May 2026	Extracts an object-centric manipulation plan from a single human video and executes it on a robot.
4	MimicFunc: Imitating Tool Manipulation from a Single Human Video via Functional Correspondence	Chao Tang et al.	CoRL 2025	Transfers manipulation skills from one human RGB-D video to robotic manipulation of novel tools.
5	UniSkill: Imitating Human Videos via Cross-Embodiment Skill Representations	Hanjung Kim et al.	CoRL 2025	Learns embodiment-agnostic skill representations from human/robot videos.
6	X-Sim: Cross-Embodiment Learning via Real-to-Sim-to-Real	Prithwish Dan et al.	CoRL 2025	Converts human RGB-D demonstrations into object-centric rewards and robot policies without teleoperation data.

Literature Survey - II

No.	Paper	Authors	Publication	Relevance
7	Phantom: Training Robots Without Robots Using Only Human Videos	Marion Lepert, Jiaying Fang, Jeannette Bohg	CoRL 2025	Generates robot-compatible training data directly from human videos and enables real-world deployment.
8	One-shot Learning for Rapid Generation of Structured Robotic Manipulation Tasks from 3D Video Demonstrations	Jaime Duque-Domingo et al.	Sep. 2025	Decomposes human demonstrations into structured primitive actions for robot execution.
9	A Neuromorphic Approach to Obstacle Avoidance in Robot Manipulation	Ahmed Abdelrahman et al.	2025	Uses SNN/event-based processing for reactive robotic-arm obstacle avoidance; demonstrates neuromorphic manipulation.
10	Designing Spiking Neural Network-Based Reinforcement Learning for 3D Robotic Arm Applications	Yuntae Park et al.	Jan. 2025	Combines SNN with TD3 for 3D robotic-arm control and reports energy-efficiency advantages.
11	An Accuracy-Improved Low-Computational-Cost Spiking Transformer Network for Robotic Arm Manipulation	M. Gao, Z. Hou, D. Zhu, C.-I. Ieong	Oct. 2025	Uses a spiking Transformer inside a diffusion-based robotic manipulation policy.
12	Spiking Neural Network-Based Energy-Efficient Framework for Real-Time Robotic Arm Manipulation	A. K. Saini et al.	Mar. 2026	Uses SNN for robotic-arm trajectory generation and inverse kinematics with physical hardware validation.

Workflow


HUMAN DEMONSTRATION
RGB VIDEO
VIDEO PRE-PROCESSING
HUMAN POSE EXTRACTION
HAND STATE EXTRACTION
OBJECT TRACKING
MOTION FEATURES
TEMPORAL SEGMENTATION
SKILL IDENTIFICATION
STRUCTURED SKILL MODEL
LEARNING FROM DEMO
SKILL GENERALISATION
ROBOT TRAJECTORY
INVERSE KINEMATICS
ROBOT EXECUTION
ROBOT STATE SIGNALS
SNN STATE ESTIMATOR
CURRENT SKILL PHASE
PHASE-AWARE EXECUTION


RGB VIDEO
REACH
APPROACH
GRASP
LIFT
TRANSPORT
PLACE
RELEASE
LEARNED MANIPULATION SKILL
ROBOT EXECUTION
SNN ESTIMATES CURRENT PHASE

Module Description

Module 1 — Video & Motion Extraction

- Input: RGB demonstration video
- Extract human pose, hand landmarks and manipulated object
- Generate frame-level motion and interaction features

Module 2 — Temporal Skill Segmentation

- Detect meaningful changes in motion and interaction
- Identify boundaries between manipulation phases
- Produce labelled segments with timestamps

Module 3 — Skill Representation & LfD

- Convert segmented demonstrations into a structured skill representation
- Learn the demonstrated movement pattern
- Generalise the skill to new object/target positions
- Future trajectory representation using DMP/ProMP-based methods

Module 4 — Robot Mapping & Execution

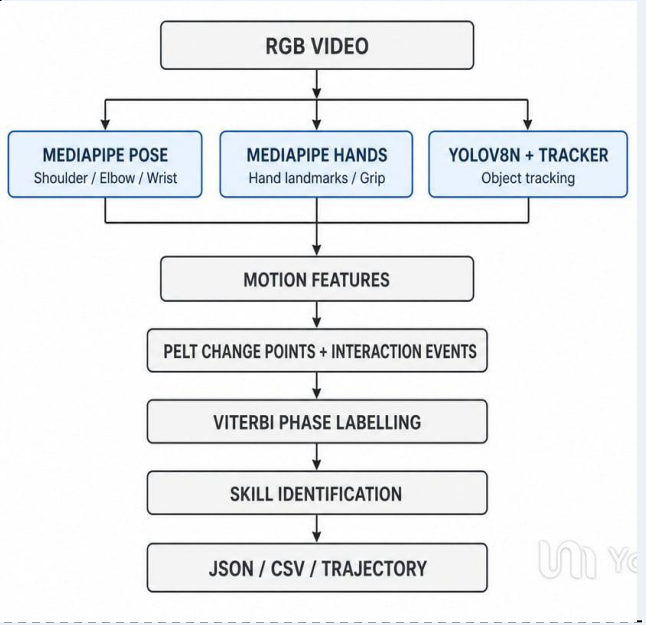
- Transform learned human motion into robot workspace
- Generate end-effector trajectory
- Apply inverse kinematics

Module 5 — SNN State Estimation

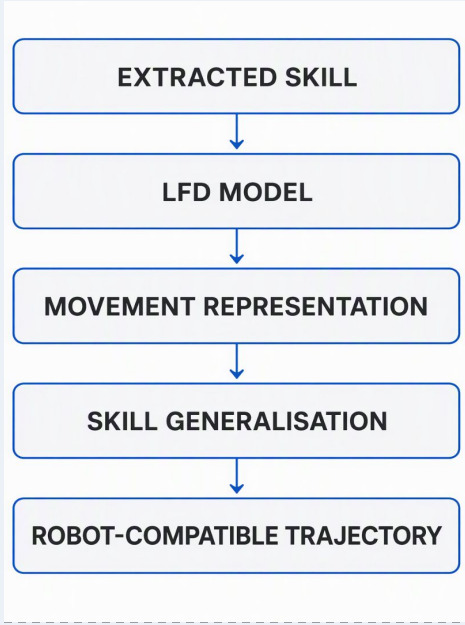
- Receive robot state information
- Learn temporal patterns of task execution

Technical Architecture

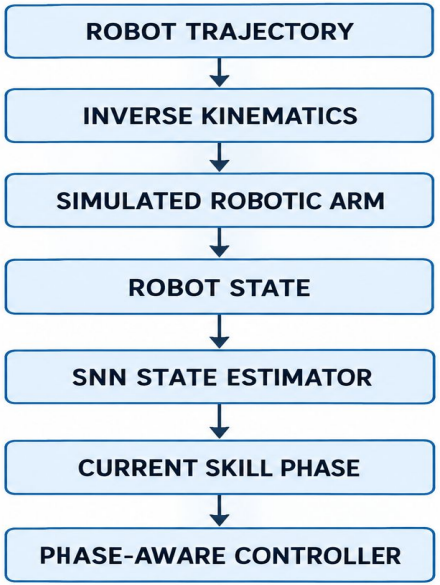
Phase 1 — Video → Skill Extraction



Phase 2 — Skill Learning



Phase 3 — Robot + SNN



Tools, Technologies, and Models Used

Current Implementation

Skill Extraction from Human Demonstration Video

Input:RGB video of a person performing a manipulation task

Current pipeline: Video → Pose/Hand/Object Extraction → Motion Features → Temporal Segmentation → Skill Identification → Structured Representation

Models currently implemented

- **MediaPipe Pose / BlazePose** — 33 body landmarks
- **MediaPipe Hands** — 21 hand landmarks
- **YOLOv8n** — object detection
- **IoU + Centroid Tracker** — object tracking
- **PELT** — statistical change-point detection
- **Viterbi decoding** — phase sequence labelling

Current evaluation

MoF: 78.8%
Edit Score: 74.7
F1@10: 81.9%
F1@50: 79.0%
Boundary MAE: 103 ms
Family-level task accuracy: 100%

Tools, Technologies, and Models Used

Video / Computer Vision

- Python
- OpenCV
- MediaPipe Pose
- MediaPipe Hands
- YOLOv8n
- NumPy
- Pandas
- SciPy
- Scikit-learn

Current Skill Extraction

- Motion-feature extraction
- PELT change-point detection
- Interaction-event detection
- Viterbi decoding
- Sequence-based skill classification

Future LfD

- Learning from Demonstration
- DMP / ProMP-based trajectory representation
- Human → Robot motion retargeting
- Inverse Kinematics
- Robotic-arm simulation

Future SNN

- Spiking Neural Network
- LIF neurons
- Temporal state/phase estimation
- Phase-aware robot control
- **snnTorch** for implementation

Hardware — Future Phase

- Raspberry Pi 5
- MG995R servos
- PCA9685 servo driver
- RGB camera
- External high-current servo power supply
- Joint-position sensing / encoders
- Gripper sensing

References

1.**Ravichandar et al. (2020)** — Recent Advances in Robot Learning from Demonstration, Annual Review of Control, Robotics, and Autonomous Systems.

2.Mandlekar et al. (2023) — MimicGen: A Data Generation System for Scalable Robot Learning using Human Demonstrations, **CoRL.**

3.Huang et al. (2023) — Inner Monologue: Embodied Reasoning through Planning with Language Models, **CoRL.**

4.Chi et al. (2025) — Diffusion Policy: Visuomotor Policy Learning via Action Diffusion, **International Journal of Robotics Research.**

5.Kim et al. (2025) — OpenVLA: An Open-Source Vision-Language-Action Model, **CoRL.**